

Struktural induction

- 1) bitstreng
- 2) Palindromer
- 3) talmaenge
- 4) traer

Bitstreng

$$\lambda$$

$$\Sigma = \{0, 1\}$$

$$\lambda \in \Sigma^*$$

$$B \in \Sigma^* \wedge b \in \Sigma \rightarrow Bb \in \Sigma^*$$

- 1) $\{\lambda\} \in \Sigma^*$
- 2) $\{0, 1\} \in \Sigma^*$

$$3) \{00, 01, 10, 11\} \in \Sigma^*$$

$$4) \{000, 001, 010, 011, 100, 101, 110, 111\} \in \Sigma^*$$

Talmangde

$$S_1 = \{3\}$$

$$S_i = S_{i-1} \cup \{x+y \mid x, y \in S_{i-1}\}$$

$$S_2 = \{3, 6\}$$

$$S_3 = \{3, 6, 9, 12\}$$

$$S = \{3n \mid n \in \mathbb{Z}^+\}$$

$$A \subseteq \mathbb{Z}$$

$$S \subseteq A$$

$$\text{Base: } S_1 \subseteq A$$

$$\text{IH: antag, at der findes } k \in \mathbb{Z}^+$$

$$S_k \subseteq A \rightarrow S_{k+1} \subseteq A$$

$$\text{IS: tag } c \in S_{k+1} \setminus S_k \text{ Da}$$

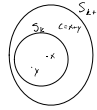
$$c = x+y$$

$$\text{hvor } x, y \in S_k$$

$$x = 3u \text{ og } y = 3v$$

$$\text{Da } c = 3u + 3v = 3(u+v)$$

$$\text{Da } c \in A$$



Palindromer

$$P_i \cup \{a, b\} \in P$$

$$B \in P \wedge b \in \Sigma \rightarrow bBb \in P$$

$$P_2 \cup \{00, 01, 10, 11\} \in P$$

traer

$$S_1 = \{\cdot\}$$

$$S_2 = \{\cdot, \Lambda\}$$

$$S_3 = \{\cdot, \Lambda, \Lambda\Lambda, \Lambda\Lambda\Lambda, \Lambda\Lambda\Lambda\Lambda\}$$

$$S = \bigcup_{i=1}^{\infty} S_i$$

serstning for et binart tree med $n(T)$ knuder

$$\text{hvor vi } n(T) \leq 2^{h(T)+1} - 1$$

Bevis

$$\text{Base } 1 = n(T) \leq 2^{h(T)+1} - 1 = 2^{1+1} - 1 = 1$$

$$\text{IH: antag at der findes } k \geq 2 \text{ sa}$$

$$n(T) \leq 2^{h(T)+1} - 1 \text{ for alle } T \in S_{k-1}$$

$$\text{IS: vis, at det gælder for alle } T \in S_k$$

$$\text{tag } S_k \setminus S_{k-1}, \text{ sa } T = T_1 \cup T_2 \text{ hvor } T_1, T_2 \in S_{k-1}$$

$$\text{fra IH}$$

$$n(T_1) \leq 2^{h(T_1)+1} - 1 \quad (\neq)$$

$$n(T_2) \leq 2^{h(T_2)+1} - 1$$

$$\text{og}$$

$$h(T) \geq 1 + h(T_1)$$

$$(\neq \neq)$$

$$h(T) \geq 1 + h(T_2)$$

$$\text{nu er}$$

$$n(T) = n(T_1) + n(T_2) + 1$$

$$(*) \leq (2^{h(T_1)+1} - 1) + (2^{h(T_2)+1} - 1) + 1$$